

Do Sanctions Inhibit Innovation? The Case of the U.S.*

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Abstract

This paper explores the impact of US sanctions on innovation in the United States. Using data from the U.S. Patent and Trademark Office (USPTO) from 2000 to 2022, we analyze patenting activity in the U.S. by inventors from sanctioned countries. The sanctions data are from the Global Sanctions Database (GSDB), which provides detailed information on sanctions by sender, target, type, and objective. Our findings show that US sanctions significantly reduce patenting activity by inventors from sanctioned countries within the US, as well as joint patents co-authored with US inventors. We also conducted a heterogeneity analysis to identify which sanction types drive these effects and performed multiple robustness checks to validate our findings. This study contributes to the growing literature on the intersection of sanctions and innovation.

Keywords: Sanctions; innovation; patenting

J.E.L. classification: F51; O3; O31

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1 Introduction

The prevalence of international sanctions has increased dramatically, with approximately ten times more active sanctions per year now than in 1950 (Morgan, Syropoulos, and Yotov 2023). Typically, sanctions are imposed to force the target country to change its “behavior”. Such changes can include policy change, the resolution of territorial disputes, the prevention and end of wars, the change of regimes, and other objectives. Research has shown that international sanctions have negative impacts on multiple aspects and macroeconomic performance of target countries. For example, Felbermayr et al. 2020a and Gutmann, Neuenkirch, and Neumeier 2023 show that sanctions decrease international trade; sanctions also cause a decrease in international capital flows (Besedeš, Goldbach, and Nitsch 2017), foreign direct investment (Biglaiser and Lektzian 2011, Mirkina 2018), and innovation (Hovhannisyan and Miromanova 2025).

However, can sanctions imposed by a country have negative effects on the sanctioning country itself? We answer this question in our study by analyzing the impact of the US sanctions on innovation in the US. We use data from the US Patent and Trademark Office (USPTO) and Global Sanctions Database (GSDB), spanning the period of 2000-2022, to disentangle the heterogeneous impacts of sanctions imposed by the US on patenting activity within the US. We use fractional patent counts as our measure of innovation, as is customary in the literature. Our measure of sanctions comes from version 3 of the Global Sanctions Database (GSDB), created and described in Syropoulos et al. (2024), Felbermayr et al. (2020b), and Kirilakha et al. (2021).¹ The wealth of sanctions data from GSDB allows us to analyze various dimensions of sanctions, including their type (arms embargoes, military, trade, financial, etc.), objective of sanctions, and the outcome. We use a two-way fixed effects model (TWFE) and Pseudo-Poisson Maximum Likelihood (PPML).

Our findings show that US sanctions significantly decrease patenting activity by inventors from sanctioned countries’ in the US. We estimate that stronger sanctions are associated with a decline of 38% in patents applied for by inventors from countries sanctioned by the US. We also find that sanctions decrease collaboration between inventors from the US and inventors from countries sanctioned by the US: the joint patent fractional count falls by about 59% after sanctions. The economic magnitudes of our estimates are substantial. Following the imposition of stringent sanctions, overall patenting in the United States by inventors from sanctioned countries declines by approximately 246 patents, while jointly authored patents between sanctioned-country inventors and U.S. co-inventors fall by roughly 17 patents. These results are robust to a number of sensitivity analyzes, including varying samples, different estimators, and various definitions of sanctions as independent variables.

While sanctions generally impact innovation, there are certain characteristics of sanctions that drive

¹GSDB provides coverage of 1,325 publicly traceable, multilateral, plurilateral, and purely bilateral sanction cases during the 1950-2022 time period, and can be accessed via <https://www.globalsanctionsdatabase.com/>.

this impact, namely the duration and number of restrictions in place. Sanctions that last at least three years and have more than three sanction types in place simultaneously are the most impactful. Disentangling the effects of sanctions by type, we find that arms and military sanctions, which restrict the movement of weapons across borders and suspend military aid, have the largest negative impact on patenting within the US by inventors from countries sanctioned by the US. This is likely due to the nature of the sanctions applied to the defense sectors. They tend to be more strenuous and have stricter enforcement. One puzzling result that emerged from the heterogeneity analysis is that travel sanctions increase overall and joint innovation, despite travel sanctions that include visa restrictions. We provide evidence on travel sanctions that encourage innovation in the sending country due to the substitution of in-person collaboration with virtual collaboration. We find that this effect is especially prominent after the increase in the use of virtual communication following the COVID-19 pandemic.

Our findings also highlight the heterogeneity in patent responses based on the average level of patenting by inventors from other countries within the US. Sanctions erode overall and joint patenting more for inventors from countries that innovate more in the US than for countries that innovate less. We hypothesize that more innovative countries are more integrated in the US patenting ecosystem and sanctions disturb this integration. We also find that failed sanctions have a more detrimental impact on overall and joint innovation. This could be because once sanctions fail, more serious diplomatic or military actions are taken against the target, which would erode established innovation networks. Finally, we address the dynamic effects of sanctions and find that it takes 4-5 years for sanctions to start having a negative impact on overall and joint innovation.

This paper makes a significant contribution to the expanding body of research on the complex relationship between sanctions and innovation, offering new insights into their broader economic implications. Our analysis is closest to two studies. The first study is by Wang et al. (2025), who use panel data from 119 countries to examine the impact of international sanctions on innovation in sender countries. They show that countries that impose (“send”) sanctions experience a significant decline in domestic patent and trademark applications. Although the question explored in their study is similar to ours, we focus on innovation of the biggest sender of sanctions, the US. We also use different methods in this study, the TWFE model estimated using PPML to account for the zero-truncated, count nature of fractional patent data, while Wang et al. (2025) use a linear estimator. Despite the differences in methodology, our findings are in line with Wang et al. (2025): sanctions decrease innovation. The second study is Cao et al. (2024). They focus on US export restrictions and technology licensing on the innovation of Chinese firms. The findings highlight that patenting by Chinese firms decreases, primarily due to decreased collaboration with US inventors. Sanctions lead to a decline in the patent output of both Chinese firms with US collaborators and US firms with Chinese collaborators. Our findings are in line with those in Cao et al. (2024): we also find that joint innovation between collaborators from the

US and sanctioned countries declines in the aftermath of sanctions.

Our paper is also related to the literature on the impacts of sanctions on innovation in target countries. For example, Wen, Zhao, and Chang (2024) conducts a cross-country study of the effects of sanctions on target countries. Wen et al. (2024) and Fu et al. (2023) use data from Chinese firms to explore the impact of sanctions on firm-level innovation. Makkonen and Mitze (2023) explore the decrease in scientific co-publication rates between the EU and Russia after the 2014 sanctions imposed on Russia in the aftermath of the Crimean conflict. Hovhannisyan and Miromanova (2025) explore the impacts of sanctions on innovation in Russia.

This study highlights a new avenue of the impact of sanctions: their potential to have a detrimental impact on sender countries which has substantial policy implications.

The remainder of the paper is organized as follows. Section 2 establishes methodological approaches to estimating the effects of sanctions on patenting in the sender countries. Section 3 describes the data sources used in the empirical analysis and provides stylized facts. Section 4 presents and discusses the estimation results, including robustness and heterogeneity analysis. Finally, Section 5 concludes.

2 Methodology

Our goal is to examine how the imposition of sanctions by the US on countries impacts those countries' innovation in the US. We examine overall and joint innovation separately. To test our hypothesis, we use the two-way fixed effects model (TWFE), presented in equation (1) estimated using the Poisson-Pseudo Maximum Likelihood (PPML) estimator. We choose the PPML estimator instead of OLS due to the count nature of the dependent variable.

$$y_{jt} = \exp(\alpha + \beta_1 US\ sanction_{jt} + X_{jt} + \gamma_j + \gamma_t + \epsilon_{jt}) \quad (1)$$

where j represents the inventors of that country and t represents year. The dependent variable y_{jt} is fractional US patent count of inventors from country j at time t , and represents either overall or joint patents, $y_{jt} \in (OVERALLpatent_{jt}, JOINTpatent_{jt})$. Overall fractional patent count refers to the total fractional count of patent applications by inventors from country j in year t . We calculate it in the following way: for every patent, we assign fractional ownership (for example, if the patent has 3 inventors, one from the US, one from Russia and one from Germany, each inventor is assigned $\frac{1}{3}$ of the ownership). We calculate this fractional ownership for every patent in our sample and then aggregate the counts by country-year level. Joint fractional patent count is calculated in a similar fashion, but we only look at patents that have at least one US inventor *and* at least one non-US inventor. We use fractional patent counts as a proxy for innovation following the established convention in the literature.

To capture the effect of sanctions on innovation, we begin with a simple dummy variable $US\ sanctions_{jt}$, which takes the value of 1 if country j is sanctioned by the US at time t , and 0 otherwise. We also experiment with using indicator variables for particular sanctions types (for example, arms embargoes, travel or financial restrictions).

The variable of interest in our analysis varies at the country-year level; therefore, we cannot include country-year fixed effects. Instead, to control for other factors that could affect innovation, we include a vector of country-year controls, X_{jt} . The vector includes GDP per capita, the presence of a diplomatic disagreement between country j and the US, the WTO and EU membership status for country j , mean years of schooling in country j , exports from the US to country j , imports from country j to the US, and FDI openness of country j .² We also control for the strictness of travel restrictions, imposed by countries after the onset of the COVID-19 pandemic in March 2020. If no restrictions are in place, the variable takes the value of 0; otherwise it's the maximum number of restrictions imposed in that year. γ_j and γ_t represent country and year fixed effects, respectively. These fixed effects control for unobserved country-level variation, and for time-specific shocks common for inventors from all countries (onset of COVID-19, macroeconomic shocks).

3 Data and Stylized Facts

We obtain innovation data from the US Patent and Trademark Office (USPTO).³ We use applications for useful invention patents that are eventually granted as our measure of innovation. We match each patent with its inventor and the inventor's residence country. This allows us to calculate fractional overall patent counts and joint patent counts, defined as all patents that have at least one US inventor and one non-US inventor. We aggregate fractional patents by country-year level, and explicitly include zero patents, i.e., if there are no patents by inventor from country j in year t , we record that observation as 0. Our full sample consists of 4,485 observations: there are 3,168 non-zero overall fractional patents observations, and 2,414 non-zero joint patent counts. Because we explicitly include implicit zeros, we end up with a strongly balanced panel of 195 countries observed between 2000 and 2022. We truncate the patent data in year 2022 because there are significantly fewer observations available for 2023 and onward due to the lag between the application and granting of patents.

Data for our variable of interest are sourced from version 3 of the Global Sanctions Database (GSDB).⁴ GSDB provides a wealth of information on 1,574 sanctions. The information in GSDB includes the type of sanction in place (arms sanction, military assistance sanction, travel sanctions, financial sanction, trade sanction, and other), the objectives of sanction (GSDB distinguishes between (1) security

² $FDI\ openness = \frac{FDI\ inflow_{jt} + FDI\ outflow_{jt}}{real\ GDP_{jt}} \times 100\%$

³The patent data used in this study is publicly available at <https://data.uspto.gov/home>.

⁴Available on request at <https://www.globalsanctionsdatabase.com/>.

related issues such as ending war, preventing war, or curbing terrorism, and (2) non-security related issues such as human rights, specific policy changes, territorial conflict, destabilizing the regime). For each sanction objective, there is a corresponding outcome: success (partial or total), failure, negotiation settlement, or ongoing.

Our independent variable of interest, $US\ sanction_{jt}$, is a dummy variable that indicates whether the country j is sanctioned by the US in year t . Since the GSDB reports the sanction by type, we interpret the variable $US\ sanction_{jt}$ as 1 if there is any type of sanction in place (at least one) and zero otherwise. However, the United States applies sanctions frequently and liberally: 43% of the observations in our sample are sanctioned, and 162 out of 195 countries had some sanctioning measure imposed. This implies that our independent variable masks a lot of heterogeneity, mixing strenuous sanctions with lighter cases. To mitigate this, we modify the independent variable in the following way. First, we focus on sanction cases that have at least three different sanction types in place (for example, trade and travel, or financial and arms sanctions). More than half of the cases in our sample have only one sanction type in place. We argue that these sanctions are less likely to be detrimental to innovation because typically they are imposed in response to minor transgressions and are typically targeted to individuals or entities. Cases where more than one type of sanctions are less prevalent and, we argue, are more stringent, therefore, they are more likely to have an effect on patenting activity. We also restrict the sanctions by duration: we recode $US\ sanction_{jt}$ as zero if a sanctioning episode lasts less than 3 years. This reduces the number of sanctioned observations to 284, and the number of sanctioned countries to 29.

We also use a comprehensive set of controls to account for country-year variation, including real GDP per capita, bilateral imports and exports between country j and the US, bilateral FDI inflows and outflows, trade gravity model controls, average education, and COVID-19 restrictions.

We obtain the gravity of trade controls from the CEPII Gravity data set (Conte, Cotterlaz, and Mayer 2022), including whether country j is a member of the World Trade Organization or the EU, as well as whether there is a diplomatic disagreement between country j and the US. We control for country fixed effects in our specifications, so we exclude gravity controls that do not vary over time. Population and real GDP are sourced from the World Bank Development Indicators, available publicly (World Bank 2024). Data on the average number of years of education are sourced from Global Data Lab, which compiles data from a variety of national and international sources, including demographic and health surveys.⁵ Biannual import and export data between the United States and the partner country j are sourced from the BACI: International Trade Database at Product-Level Data (Gaulier and Zignago 2010). The bilateral FDI inflow and outflow data is described in Steenbergen et al. (2022) and was shared with us by the authors of the data set at the World Bank. FDI control captures whether the US sends (outflow) or receives (inflow) FDI from the partner country j in year t . The bilateral FDI

⁵Mean years of schooling by country are available via <https://globaldatalab.org/shdi/metadata/msch/>.

data have many missing values, so while we use the FDI as a control in a robustness check exercise, in our preferred specifications, we eliminate it to preserve as many observations as possible. We source data on COVID-19 restriction from the data set “International travel controls during the COVID-19 pandemic”, compiled by the Blavatnik School of Government, University of Oxford (Blavatnik School of Government, University of Oxford 2023), with minor processing by Our World in Data.⁶ This data set contains information on daily travel restrictions during the COVID-19 pandemic. The restrictions are coded as a categorical variable, which varies between 0 (no restrictions) and 4 (ban on all regions or total border closure), with screening and quarantine arrivals as milder measures. Given that we use the monthly data for the patent counts, we collapse the data to the maximum number of restrictions within a year.⁷

After combining all the data sources, we obtain a strongly balanced panel dataset of 195 countries, observed annually between 2000 and 2022. In all specifications, we exclude the US inventors and inventors from China,⁸ which effectively reduces the number of observations to 4439. The summary statistics (see Tables 1 and 2) and stylized facts about the data are presented next.

Table 1: Summary Statistics by Sanction Status

	Non-sanctioned countries	Sanctioned countries	Pooled sample
Overall fractional patent count	742.12 (3937.60)	100.65 (611.66)	649.06 (3655.14)
Joint fractional patent count	30.03 (105.34)	22.23 (140.11)	28.90 (111.08)
Real GDP/capita (USD)	16842.21 (24120.72)	3053.50 (2713.33)	14964.80 (22933.42)
Ave. education (years)	8.48 (3.24)	6.27 (2.79)	8.15 (3.27)
Imports (mln USD)	9.53 (34.76)	3.80 (10.23)	8.71 (32.48)
Exports (mln USD)	6.70 (23.15)	1.39 (4.43)	5.92 (21.52)
Observations	667	3,818	4439

Non-sanctioned countries patent more on average and have higher counts for joint innovation with the US. They also tend to be richer, more educated, and trade more with the US (Table 1).

We also provide summary statistics by country’s level of innovation in Table 2. We separate all countries into quartiles based on the median overall patent count of country j throughout the entire sample.⁹ Each country j belongs to one quartile and does not switch quartiles throughout the sample

⁶Our World in Data is accessible via <https://ourworldindata.org/>. The website publishes research and data on socially relevant topics, such as poverty, inequality, health, etc.

⁷We also experiment with using the average and median number of restrictions, and the results are robust to this change.

⁸China’s invention grows steadily over time, as shown in Figure 1, but GSDB classifies it as a country sanctioned by the US during the whole sample period, and we exclude China to capture the true effect of sanctions.

⁹We experiment with using the maximum overall patent count, as well as joint innovation to determine quartiles, and the results remain consistent.

Table 2: Summary Statistics by Innovation

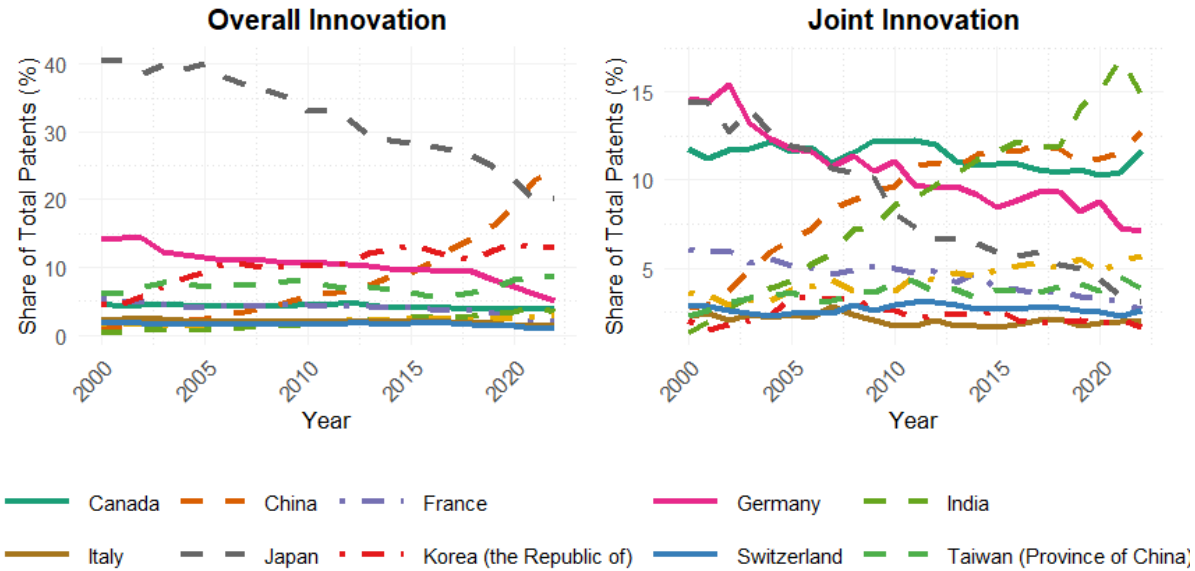
	Bottom 25%	Q2	Q3	Top 25%	Pooled
Overall fractional patent count	0.23 (0.70)	1.33 (1.89)	13.10 (16.45)	2708.04 (7107.94)	649.06 (3655.14)
Joint fractional patent count	0.03 (0.16)	0.21 (0.45)	1.63 (2.36)	119.33 (202.61)	28.90 (111.08)
Real GDP/capita (USD)	5592.99 (9937.32)	8125.95 (15292.70)	18855.98 (30771.91)	27953.63 (23248.73)	14964.80 (22933.42)
Ave. education (years)	5.94 (2.82)	6.66 (2.86)	9.34 (2.38)	10.66 (2.32)	8.15 (3.27)
Imports (mln USD)	0.35 (1.26)	1.36 (2.88)	3.49 (8.79)	30.67 (59.84)	8.71 (32.48)
Exports (mln USD)	0.16 (0.28)	0.85 (1.37)	2.33 (3.47)	21.16 (39.81)	5.92 (21.52)
Observations	1,541	713	1,127	1,104	4439

period. The list of countries by quartile is presented in Table A1. The bottom three quartiles patent significantly less (overall and joint) than countries in the top 25%. Using correlation analysis, we conclude that as countries become more educated, richer, and trade more with the US, the number of overall and joint patents filed in the US increases.

Figure 1: Top 10 Countries by Innovation Over Time

Share of total innovation

Top 10 countries (USA excluded)



Note: Shares calculated as country's fractional patents as % of total

Notes: Top 10 foreign countries whose inventors contribute the most to the patenting within the US. Inventors from the US are excluded from the data.

We visualize the top 10 countries patenting the most in the US (overall and joint) in Figure 1. The

share of overall and joint fractional patents that belong to the Japanese inventors has fallen during the sample duration. However, Chinese inventors are patenting more in general and in collaboration with US inventors. Among the top 10 countries are India, Canada, Germany, and Israel.

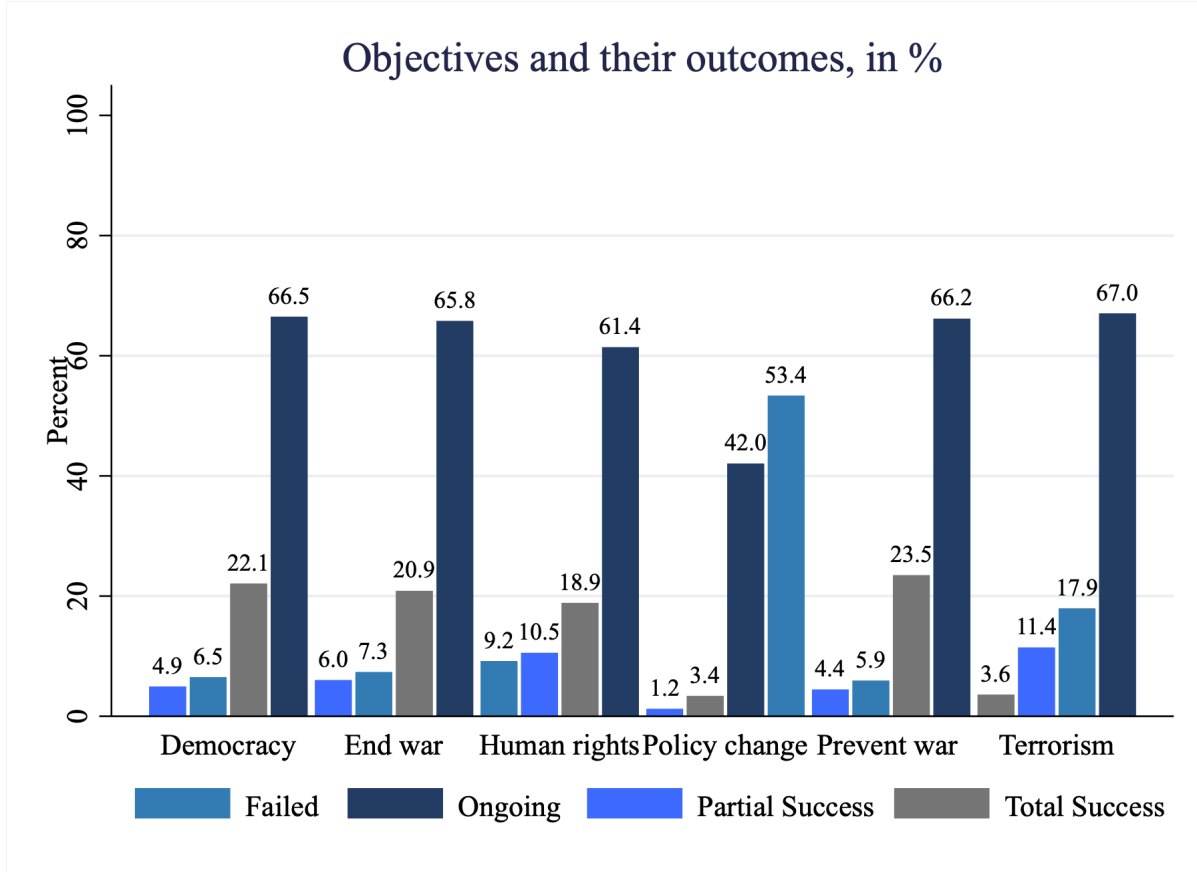
Table 3: Sanction Outcomes by Objective

Objective	Failed	Negotiation	Ongoing	Partial Success	Total Success	Total
Policy change	745 (53.37%)	0 (0.00%)	587 (42.05%)	17 (1.22%)	47 (3.37%)	1,396 (100%)
Human rights	74 (9.05%)	12 (1.47%)	495 (60.51%)	85 (10.39%)	152 (18.58%)	818 (100%)
End war	44 (7.27%)	6 (0.99%)	394 (65.12%)	36 (5.95%)	125 (20.66%)	605 (100%)
Prevent war	28 (5.77%)	12 (2.47%)	313 (64.54%)	21 (4.33%)	111 (22.89%)	485 (100%)
Terrorism	80 (17.86%)	2 (0.45%)	299 (66.74%)	51 (11.38%)	16 (3.57%)	448 (100%)
Democracy	25 (6.46%)	2 (0.52%)	256 (66.15%)	19 (4.91%)	85 (21.96%)	387 (100%)
Other	19 (5.14%)	1 (0.27%)	266 (71.89%)	0 (0.00%)	84 (22.70%)	370 (100%)
Destabilize regime	61 (32.11%)	16 (8.42%)	110 (57.89%)	0 (0.00%)	3 (1.58%)	190 (100%)
Territorial conflict	25 (32.05%)	0 (0.00%)	30 (38.46%)	23 (29.49%)	0 (0.00%)	78 (100%)
Total	1,101 (23.05%)	51 (1.07%)	2,750 (57.57%)	252 (5.28%)	623 (13.04%)	4,777 (100%)

Notes: Sorted by the sanction objective frequency.

Table 3 shows the breakdown of the sanction objective and their outcomes. Each sanction episode can have multiple objectives; therefore, the total number of objectives in Table 3 is almost as large as the number of observations in the sample, and larger than the number of observations with stringent sanction in place. The most commonly used objective is the change of a specific policy, followed by the protection or reinstatement of human rights, and ending war. To visualize how successful the sanctions are, we plot the top 5 objectives and outcomes in Figure 2. Most often, the objectives of the sanctions either fail or stay in place for a prolonged period of time. Note that this does not speak to the general success of sanctions because each sanctioning episode can have multiple objectives (up to 20), and they do not resolve in the same outcome.

Figure 2: Distribution of sanction cases by objectives and outcomes



Notes: 5 most commonly used sanction objectives are ordered alphabetically. The negotiation settlement outcome is removed in the interest of space, because this outcome is very rare.

We also explore the interplay between sanction types within each sanction episode. First, in Table 4 we provide the breakdown of observations in our sample by sanction strength, which we calculate by counting how many distinct sanction types are in place at once. There are significantly more sanctioning episodes with one sanction type in place, which is why we restrict our analysis to what we call "strenuous" sanctions (sanctions with at least 3 sanction types in place).¹⁰

Table 4: Sanction Outcomes by Objective

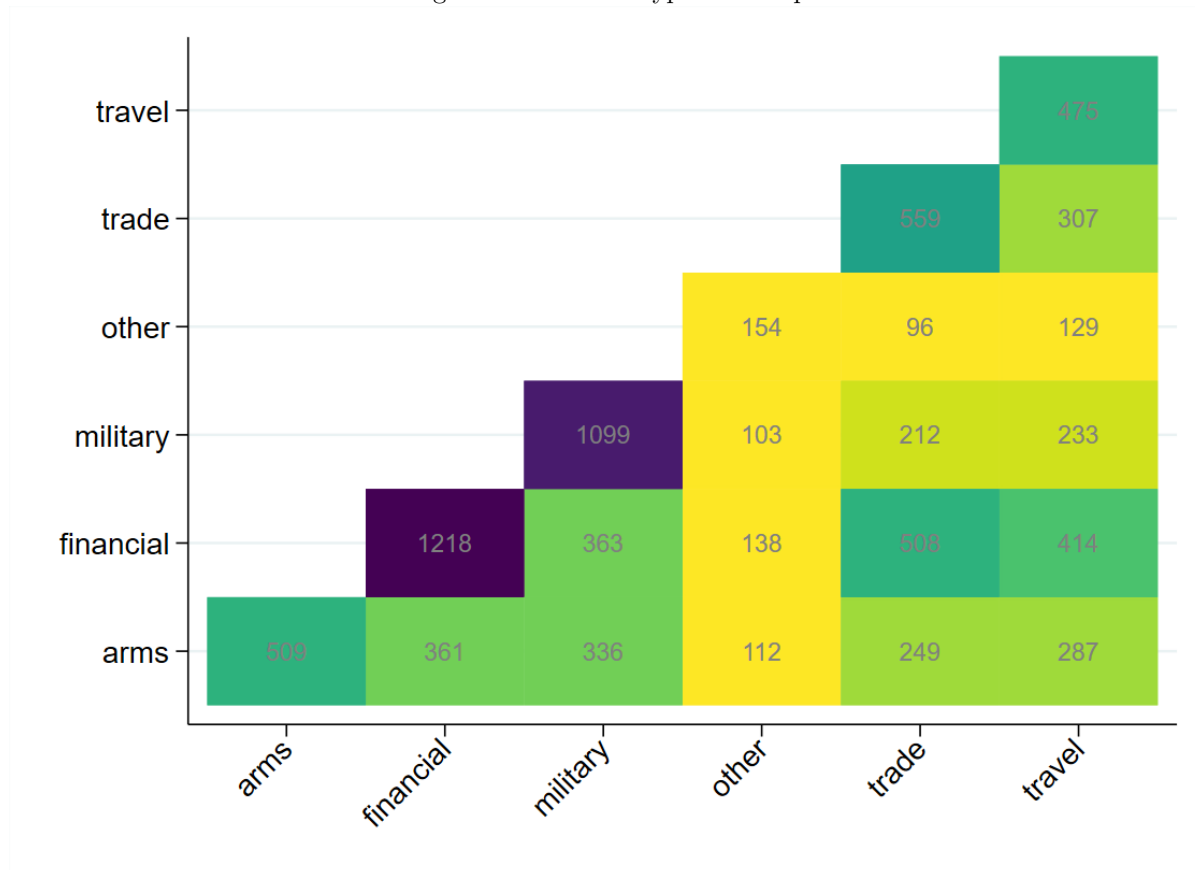
Sanctions strength	Freq.	Percent
0	2,387	53.22
1	1,227	27.36
2	393	8.76
3	177	3.95
4	89	1.98
5	158	3.52
6	54	1.20
Total	4,485	100.00

Finally, to understand what types of sanction are more prevalent, we analyze the frequency of occurrence for each type, as well as the frequency with which different types of sanction are in place

¹⁰The results for minimum 2 and minimum 3 types in place are very similar, so we stick with minimum 2 types.

together in the same sanction episode. Financial and military sanctions are imposed more frequently than others. Other types of sanctions include a variety of measures (including diplomatic restrictions) and are imposed less frequently. Figure 3 shows a heatmap of sanction types. Along the 45 degree line (i.e., trade-trade), the counts of individual sanction types are captured. We can see from the figure that financial sanctions tend to be imposed together with trade or travel sanctions, but there are no other systematic relationships that jump out. We do not observe a strong correlation between the sanction types within the sanctioning episodes.

Figure 3: Sanction Type Heatmap



Notes: Sanction types are labeled on the x and y axes. Each hexagon represents a pair of sanction types. The darker the color, the more frequent the combination of sanction types is. The hexagon labels show the actual count of observations that have the two sanction types in place. We use the full sample for the heatmap, instead of the strenuous sanction sample used in the estimations.

4 Results

4.1 Benchmark Results and Robustness Checks

We begin our analysis by estimating the equation (1). The results for overall innovation as a dependent variable are presented in Table 5, and for joint innovation as a dependent variable are in Table 6. We include the controls in the vector X_{jy} one by one in the columns of Tables 5 and 6, following Hovhannisyan and Keller (2015).

Table 5: Benchmark results: Overall innovation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
US sanction _{jt}	-0.83*** (0.21)	-0.80*** (0.19)	-0.80*** (0.20)	-0.57*** (0.12)	-0.57*** (0.12)	-0.48*** (0.17)	-0.47*** (0.18)
GDP per capita _{jt}	1.58*** (0.49)	1.52*** (0.47)	1.52*** (0.48)	2.02*** (0.32)	2.01*** (0.31)	1.82*** (0.33)	1.74*** (0.30)
Dipl disagree _{jt}		-0.19* (0.11)	-0.18* (0.11)	-0.10* (0.05)	-0.10* (0.05)	-0.10** (0.05)	-0.08 (0.05)
WTO member _{jt}			0.03 (0.17)	0.52 (0.36)	0.52 (0.36)	0.46 (0.34)	0.45 (0.28)
EU member _{jt}			0.02 (0.10)	-0.02 (0.07)	-0.02 (0.07)	-0.01 (0.08)	-0.06 (0.07)
Ave edu _{jt}				0.12* (0.07)	0.12* (0.07)	0.11 (0.08)	0.10 (0.08)
COV19 restr _{jt}					-0.004 (0.05)	0.006 (0.04)	0.026 (0.05)
Log import _{jt}						-0.02 (0.13)	0.05 (0.13)
Log export _{jt}						0.17* (0.10)	0.15 (0.09)
FDI openness _{jt}							-122.20 (198.50)
Constant	-7.03 (5.03)	-6.15 (4.90)	-6.12 (5.00)	-13.44*** (2.90)	-13.43*** (2.84)	-13.87*** (2.83)	-13.73*** (2.81)
Observations	4,146	4,146	4,146	3,839	3,839	3,813	2,672

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t). Dependent variable is overall fractional patent count.

We find that strenuous sanctions reduce overall and joint patenting. The coefficient on our main variable of interest, *US sanction_{jt}* is negative and highly statistically significant for overall fractional patent counts (see Table 5) and the joint fractional patent counts (see Table 6).

The country-year control variables have expected signs in all specifications. Our preferred benchmark specification is presented in column (6) of both tables, and includes the majority of the country-year controls, with the exception of the FDI openness. FDI data has many missing observations, losing more than a thousand observations between columns (6) and (7) in both tables. We interpret the results in column (6) in Tables 5 and 6 as follows:

- *Overall innovation*: $\exp^\beta - 1 = \exp^{(-0.48)} - 1 \approx -38\%$. To put this finding in context, we use the average fractional overall patent count from Table 1 to estimate that after the imposition of strenuous sanctions overall patenting in the US by inventors from sanctioned countries declines by about 246 patents.
- *Joint innovation*: $\exp^\beta - 1 = \exp^{(-0.88)} - 1 \approx -59\%$. Using the same methodology as for the overall patent count, this translates to a decline of about 17 patents jointly filed by the inventors from sanctioned countries with the US inventors after sanctions are imposed.

Table 6: Benchmark effects: Joint innovation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
US sanction _{jt}	-1.11*** (0.12)	-1.10*** (0.12)	-1.09*** (0.12)	-1.08*** (0.12)	-1.09*** (0.11)	-0.88*** (0.15)	-0.70*** (0.15)
GDP per capita _{jt}	1.53*** (0.36)	1.52*** (0.36)	1.54*** (0.37)	1.47*** (0.32)	1.47*** (0.32)	1.07*** (0.20)	1.11*** (0.20)
Dipl disagree _{jt}		-0.04 (0.06)	-0.04 (0.06)	-0.05 (0.06)	-0.05 (0.06)	-0.06 (0.06)	-0.04 (0.06)
WTO member _{jt}			-0.02 (0.17)	-0.02 (0.15)	-0.02 (0.15)	-0.06 (0.12)	-0.02 (0.10)
EU member _{jt}			-0.11 (0.07)	-0.08 (0.07)	-0.07 (0.06)	-0.09 (0.06)	-0.11* (0.07)
Ave edu _{jt}				0.15** (0.06)	0.15** (0.06)	0.13** (0.06)	0.12* (0.06)
COV19 restr _{jt}					-0.013 (0.07)	0.017 (0.06)	0.039 (0.05)
Log import _{jt}						0.14 (0.13)	0.13 (0.13)
Log export _{jt}						0.20** (0.08)	0.22*** (0.08)
FDI openness _{jt}							-17.30 (102.00)
Constant	-9.82*** (3.68)	-9.67*** (3.69)	-9.86*** (3.70)	-10.79*** (3.23)	-10.78*** (3.22)	-12.37*** (3.44)	-12.94*** (3.47)
Observations	3,652	3,652	3,652	3,433	3,433	3,407	2,384

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t). Dependent variable is joint fractional patent count.

Next, we test the robustness of our findings. First, we use a different definition of our independent variable of interest. In our benchmark results, we focus on strenuous US sanctions, which we define as sanction episodes that have at least 3 sanction types in place and last at least 3 years. In columns (1) and (2) of Table 7, we replace the variable *US sanction_{jt}* in our equation (1) with the variable *Full US sanction_{jt}*. Unlike the originally defined *US sanction_{jt}* variable, *Full US sanction_{jt}* includes what we call “weaker sanctions”, along with the strenuous sanctions (essentially, it is a broader definition of sanctions). It is also an indicator variable and has a value of 1 if the US imposes strenuous *or* weak sanctions on country *j* in year *t* and 0 otherwise. Although the coefficients are negative for the overall and joint patent counts when we use a broader definition of sanctions, there is no statistical significance. We conclude that while sanctions impact innovation, there are certain characteristics of sanctions that drive this impact, namely the duration and number of restrictions in place.

Table 7: Robustness Checks: Alternative Independent Variables

	Full sanctions		Strength		Lagged US Sanction		Lagged strength		Pre-COVID-19 sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Overall	Joint	Overall	Joint	Overall	Joint	Overall	Joint	Overall	Joint
Main variables										
Full US sanction _{jt}	-0.082 (0.055)	-0.100 (0.102)								
Sanctions strength _{jt}			-0.053 (0.034)	-0.099** (0.044)						
US sanction _{jt-1}					-0.283* (0.156)	-0.718*** (0.138)				
Sanctions strength _{jt-1}							-0.071* (0.041)	-0.094** (0.046)		
US sanction _{jt}									-0.454** (0.182)	-0.868*** (0.150)
Control variables										
GDP per capita _{jt}	1.792*** (0.319)	1.088*** (0.210)	1.791*** (0.330)	0.981*** (0.202)	1.744*** (0.304)	1.057*** (0.202)	1.689*** (0.309)	0.969*** (0.208)	1.691*** (0.349)	1.057*** (0.205)
Dipl disagree _{jt}	-0.092* (0.048)	-0.067 (0.053)	-0.096* (0.052)	-0.067 (0.048)	-0.090 (0.057)	-0.051 (0.053)	-0.085 (0.057)	-0.050 (0.049)	-0.072* (0.043)	-0.024 (0.076)
WTO member _{jt}	0.527 (0.334)	0.030 (0.181)	0.576* (0.312)	0.201 (0.169)	0.468 (0.297)	-0.036 (0.113)	0.605** (0.278)	0.191 (0.168)	0.383 (0.308)	-0.158 (0.119)
EU member _{jt}	0.001 (0.081)	-0.097* (0.058)	0.000 (0.076)	-0.072 (0.060)	-0.021 (0.077)	-0.099* (0.058)	-0.001 (0.075)	-0.080 (0.054)	0.291* (0.168)	0.173 (0.157)
Ave edu _{jt}	0.129 (0.083)	0.141** (0.064)	0.124 (0.081)	0.142** (0.063)	0.129 (0.087)	0.135** (0.066)	0.145* (0.085)	0.142** (0.063)	0.061 (0.068)	0.108* (0.061)
COV19 restr _{jt}	0.012 (0.043)	0.027 (0.058)	0.008 (0.043)	0.018 (0.060)	0.005 (0.040)	0.015 (0.057)	0.007 (0.044)	0.017 (0.059)	-	-
Log import _{jt}	-0.057 (0.125)	0.119 (0.100)	-0.049 (0.120)	0.095 (0.107)	-0.002 (0.136)	0.121 (0.127)	-0.046 (0.124)	0.079 (0.105)	0.031 (0.133)	0.137 (0.128)
Log export _{jt}	0.188* (0.104)	0.248** (0.103)	0.173* (0.100)	0.219** (0.098)	0.166* (0.095)	0.175** (0.081)	0.168* (0.097)	0.192** (0.097)	0.170* (0.087)	0.234*** (0.073)
Constant	-13.49*** (2.732)	-13.24*** (3.216)	-13.35*** (2.775)	-11.44*** (2.611)	-13.55*** (2.868)	-11.61*** (3.475)	-12.57*** (2.749)	-10.56*** (2.505)	-12.82*** (3.138)	-12.67*** (3.333)
Observations	3,813	3,407	3,813	3,407	3,659	3,247	3,659	3,247	3,309	2,942

In the second robustness check, we replace the variable *US sanction_{jt}* in the equation (1) with the variable *Sanctions strength_{jt}*. This variable is constructed by adding individual dummies for each sanction type (arms, military, trade, travel, financial, etc.). The sanction strength variable varies between 0 and 6, 0 for no sanction in place and 6 for all sanction types in place. We present these results in columns (3) and (4) of Table 7. The majority sanctions episodes have one sanction type in place (see Table 4), which explains the lack or lower statistical significance of the effect of sanctions. We find that stronger sanctions have a statistically significant negative impact on joint innovation, and we do not find statistically significant results for overall innovation. This is likely due to the fact that joint innovation is more complex and rare, so sanctions are more likely to have adverse effects on it.

Next, we address the possibility that sanctions could have delayed impacts on innovation due to longer adjustment periods for R&D budgets and the delay in patent applications that are eventually granted. We replace the independent variable with its lagged version. In columns (5) and (6) of Table 7, we use the variable *US sanction_{jt}* lagged by one year, and in columns (7) and (8), we use variable *Sanctions strength_{jt}* lagged by one year. We find a statistically significant impact of both lagged variables on overall and joint patent counts.

We also consider whether the COVID-19 pandemic could be driving our results, even if we control for year fixed effects and the maximum strength of the COVID-19 travel restrictions in all specifications. We restrict the sample to the years 2000 - 2019, and re-estimate the specification in equation (1) for

overall and joint innovation. These results are presented in columns (9) and (10) of Table 7. We find that the sanction effects are negative, statistically significant, and of similar magnitude as our benchmark results.

While we find that strenuous sanctions matter for innovation and exclude the “weaker sanctions” from our definition of the variable of interest $US\ sanction_{jt}$, we want to ensure that this exclusion does not bias our results. These robustness checks are documented in Table 8. In columns (1) and (4), we provide the benchmark results for ease of comparison. In columns (2) and (5), we include an additional dummy indicator for weaker sanctions, $Weaker\ US\ sanction_{jt}$, in equation (1). $Weaker\ US\ sanction_{jt}$ takes the value of 1 if weaker sanctions are in place and 0 otherwise, and the coefficient and statistical significance of the main coefficient of interest remain similar.¹¹ In columns (3) and (6), we exclude all weaker sanction observations from the sample and re-estimate equation (1) on the new sample. The magnitude of the $US\ sanction_{jt}$ coefficient is smaller for overall and joint innovation, but still statistically significant. We conclude that the exclusion of weaker sanctions from our main variable of interest does not impact our results in a meaningful way.

Table 8: Robustness Checks: Weaker Sanctions

	Overall innovation			Joint innovation		
	(1) Benchmark	(2) Ctrl for weaker sanc.	(3) Exclude weaker sanc.	(4) Benchmark	(5) Ctrl for weaker sanc.	(6) Exclude weaker sanc.
$US\ sanction_{jt}$	-0.48*** (0.17)	-0.51*** (0.16)	-0.34** (0.16)	-0.88*** (0.15)	-0.87*** (0.17)	-0.77*** (0.20)
$Weaker\ US\ sanction_{jt}$		-0.09 (0.06)			-0.03 (0.12)	
$GDP\ per\ capita_{jt}$	1.82*** (0.33)	1.80*** (0.31)	2.29*** (0.41)	1.07*** (0.20)	1.08*** (0.22)	1.35*** (0.31)
$Dipl\ disagree_{jt}$	-0.10** (0.05)	-0.09* (0.04)	-0.03 (0.05)	-0.06 (0.06)	-0.06 (0.05)	-0.03 (0.06)
$WTO\ member_{jt}$	0.46 (0.34)	0.48 (0.33)	0.12 (0.07)	-0.06 (0.12)	-0.05 (0.14)	-0.07 (0.13)
$EU\ member_{jt}$	-0.01 (0.08)	0.00 (0.08)	-0.07 (0.05)	-0.09 (0.06)	-0.09 (0.06)	-0.15** (0.06)
$Ave\ edu_{jt}$	0.11 (0.08)	0.13 (0.08)	0.04 (0.06)	0.13** (0.06)	0.14** (0.06)	0.13 (0.08)
$COV19\ restr_{jt}$	0.01 (0.04)	0.01 (0.04)	0.03 (0.03)	0.02 (0.06)	0.02 (0.05)	0.02 (0.05)
$Log\ import_{jt}$	-0.02 (0.13)	-0.06 (0.13)	0.04 (0.10)	0.14 (0.13)	0.13 (0.11)	0.21* (0.13)
$Log\ export_{jt}$	0.17* (0.10)	0.18* (0.11)	0.09 (0.09)	0.20** (0.08)	0.20** (0.09)	0.14 (0.09)
Constant	-13.87*** (2.83)	-13.43*** (2.71)	-17.24*** (2.86)	-12.37*** (3.44)	-12.56*** (3.83)	-15.28*** (4.32)
Observations	3,813	3,813	2,245	3,407	3,407	1,915

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t).

To address the concern that our results are driven by the most intense outlier cases, we perform the following three exercises. These results are recorded in Table 9. In columns (1) and (4), we exclude Cuba and the Democratic People’s Republic of Korea due to the multi-decade nature of sanctions imposed on these countries, and Iraq due to the change in the regime in 2003. In columns (2) and (5), we exclude observations with all 6 sanction types in place (strongest, or most intense, sanctions). In columns (3) and

¹¹ “Weaker sanctions” have fewer than 3 types of restrictions in place and last less than 3 years.

(6), we exclude all observations for countries that ever had all 6 sanction types in place. The coefficient of interest for overall and joint innovation maintains comparable statistical significance and magnitude.

Table 9: Robustness Checks: Outlier Sanction Intensity

	Overall innovation			Joint innovation		
	(1) Outlier ctries	(2) Excl obs	(3) Excl ctries	(4) Outlier ctries	(5) Excl obs	(6) Excl ctries
US sanction _{jt}	-0.48*** (0.17)	-0.48*** (0.17)	-0.48*** (0.17)	-0.88*** (0.15)	-0.88*** (0.15)	-0.88*** (0.15)
GDP per capita _{jt}	1.82*** (0.33)	1.82*** (0.33)	1.82*** (0.33)	1.06*** (0.20)	1.06*** (0.20)	1.06*** (0.20)
Dipl disagree _{jt}	-0.10** (0.05)	-0.10** (0.05)	-0.10** (0.05)	-0.06 (0.06)	-0.06 (0.06)	-0.06 (0.06)
WTO member _{jt}	0.46 (0.34)	0.46 (0.34)	0.47 (0.34)	-0.06 (0.12)	-0.06 (0.12)	-0.06 (0.12)
EU member _{jt}	-0.01 (0.08)	-0.01 (0.08)	-0.01 (0.08)	-0.09 (0.06)	-0.09 (0.06)	-0.09 (0.06)
Ave edu _{jt}	0.11 (0.08)	0.11 (0.08)	0.11 (0.08)	0.13** (0.06)	0.13** (0.06)	0.13** (0.06)
COV19 restr _{jt}	0.01 (0.04)	0.01 (0.04)	0.01 (0.04)	0.02 (0.06)	0.02 (0.06)	0.02 (0.06)
Log import _{jt}	-0.02 (0.13)	-0.02 (0.13)	-0.02 (0.13)	0.14 (0.13)	0.14 (0.13)	0.14 (0.13)
Log export _{jt}	0.17* (0.10)	0.17* (0.10)	0.17* (0.10)	0.20** (0.08)	0.20** (0.08)	0.20** (0.08)
Constant	-13.89*** (2.83)	-13.87*** (2.83)	-13.88*** (2.83)	-12.38*** (3.44)	-12.37*** (3.44)	-12.37*** (3.44)
Observations	3,771	3,789	3,675	3,365	3,385	3,299

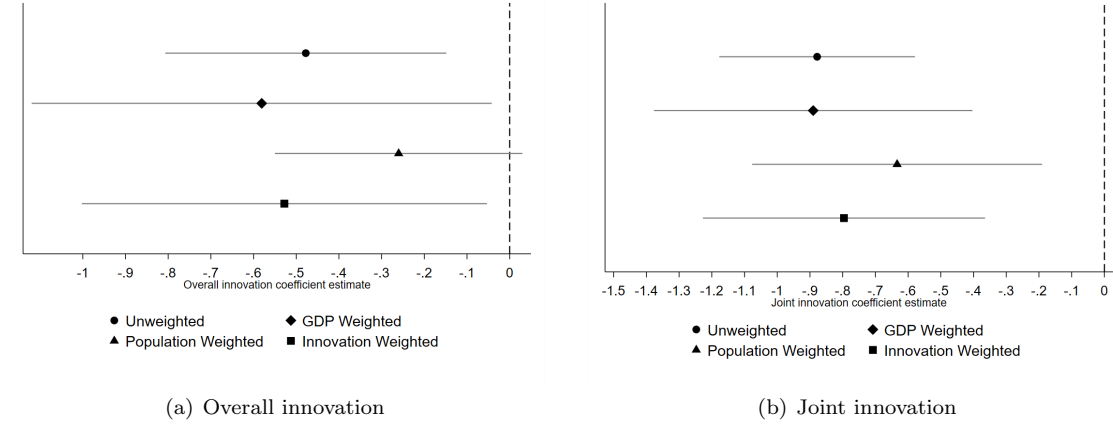
Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t). In columns (1) and (4), sample excludes Cuba, the Democratic People's Republic of Korea, and Iraq. In columns (2) and (5), sample includes all observations where Sanctions strength_{jt}=6. In columns (3) and (6), the sample excludes all countries that have Sanctions strength_{jt}=6 at some point during the sample.

As another robustness check, we re-estimate equation (1) with three different weighting schemes: real GDP, population, and median innovation. We plot the coefficients and the 95% confidence intervals for overall and joint innovation in Figure 4. Using different weighting schemes produces results similar in magnitude and significance to the benchmark estimates.

Finally, we address the dynamic effects of sanctions, as well as anticipation of sanctions. We include three leads and five lags of the variable *US sanction_{jt}* in equation (1), and re-estimate it for overall and joint patents. We plot the coefficients and the 95% confidence intervals in Figure 5. The period after imposition of sanctions, $t + 1$, is excluded for comparison. We find some evidence of overall anticipation one year before sanctions, and no anticipation for the joint patent count. We also find that it takes 4-5 years for sanctions to start having a negative impact on overall and joint innovation. This finding is in line with the timing of patent filing and the average time between filing a patent application and a patent being granted.¹² We hypothesize that anticipation for the overall patent count is most likely due

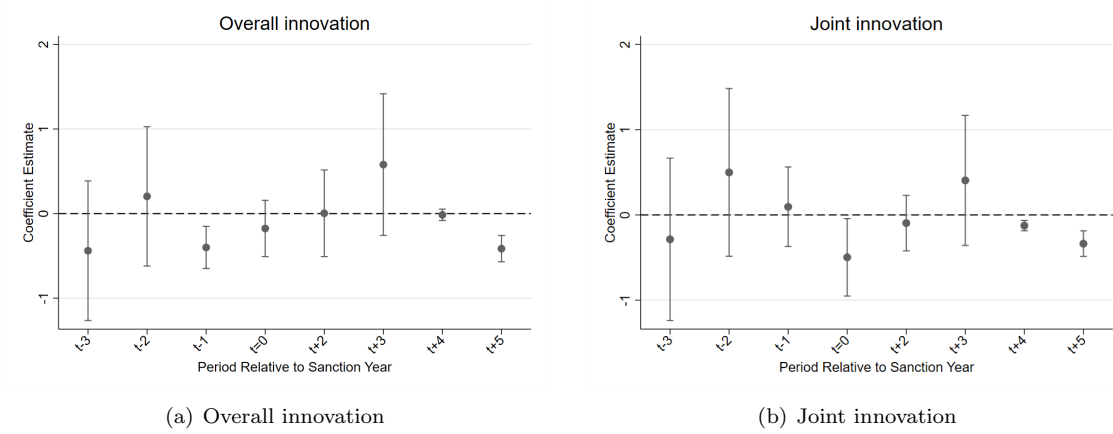
¹²USPTO reports that, on average, it takes between 28 and 33 months for the patent application to be processed. This information is available publicly at <https://www.uspto.gov/dashboard/patents/pendency.html>.

Figure 4: Robustness Check: Weighting Schemes



to the noisiness of the estimates: while there is a statistically significant negative effect one year before the sanctions, there is no statistically significant effect for several years after that.

Figure 5: Robustness Check: Dynamic Effects of US Sanctions



4.2 Heterogeneity Analysis

We begin to explore heterogeneity by analyzing how different types of sanctions affect innovation. Our main variable of interest, $US\ sanction_{jt}$, combines all types of sanctions in one, but different types of restrictions could have different impacts on innovation. To disentangle these effects, we replace $US\ sanction_{jt}$ in equation (1), with an indicator variable for each of the sanction types. For example, $Arms\ sanction_{jt}$ takes the value of 1 if arms sanction is imposed on country j in year t and 0 otherwise. If we include all types of sanctions together in the estimation equation, there is a concern that the estimates produced would be noisy. We follow Besedeš (2024) and include sanction type indicators one by one. We present these results in Table 10.

Table 10: Heterogeneity of Sanction Types

	Overall innovation						Joint innovation					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Arms sanction _{jt}	-0.50*** (0.18)						-0.80*** (0.17)					
Military sanction _{jt}		-0.27** (0.13)						-0.47*** (0.12)				
Trade sanction _{jt}			0.12 (0.30)						-0.27** (0.12)			
Financial sanction _{jt}				0.12 (0.29)						-0.27** (0.11)		
Travel sanction _{jt}					0.92** (0.37)						0.32** (0.14)	
Other sanction _{jt}						0.89* (0.45)						0.23 (0.41)
GDP per capita _{jt}	1.82*** (0.33)	1.75*** (0.37)	1.86*** (0.32)	1.86*** (0.32)	1.81*** (0.32)	1.83*** (0.33)	1.07*** (0.20)	0.82*** (0.19)	0.93*** (0.19)	0.94*** (0.18)	1.10*** (0.21)	1.10*** (0.21)
Dipl disagree _{jt}	-0.10** (0.05)	-0.11** (0.05)	-0.10** (0.05)	-0.10** (0.05)	-0.09** (0.05)	-0.10** (0.05)	-0.06 (0.06)	-0.08** (0.04)	-0.08* (0.04)	-0.08* (0.04)	-0.07 (0.05)	-0.07 (0.05)
WTO member _{jt}	0.46 (0.34)	0.47 (0.34)	0.38 (0.29)	0.38 (0.29)	-0.14 (0.24)	0.46 (0.34)	-0.06 (0.12)	-0.04 (0.12)	0.13 (0.11)	0.13 (0.11)	-0.32*** (0.11)	-0.09 (0.13)
EU member _{jt}	-0.01 (0.08)	0.00 (0.08)	-0.03 (0.07)	-0.03 (0.07)	-0.04 (0.07)	-0.02 (0.08)	-0.10 (0.06)	-0.05 (0.07)	-0.07 (0.06)	-0.07 (0.06)	-0.12* (0.06)	-0.11* (0.06)
Ave edu _{jt}	0.11 (0.08)	0.11 (0.08)	0.11 (0.08)	0.11 (0.08)	0.10 (0.07)	0.11 (0.08)	0.13** (0.06)	0.12* (0.06)	0.12* (0.06)	0.12* (0.06)	0.13* (0.06)	0.13** (0.07)
COVID19 restr _{jt}	0.01 (0.04)	0.00 (0.04)	0.01 (0.04)	0.01 (0.04)	0.02 (0.04)	0.01 (0.04)	0.02 (0.06)	0.01 (0.06)	0.01 (0.06)	0.01 (0.06)	0.03 (0.06)	0.02 (0.06)
Log import _{jt}	-0.01 (0.13)	-0.02 (0.13)	-0.01 (0.12)	-0.01 (0.12)	0.03 (0.11)	-0.01 (0.13)	0.14 (0.13)	0.11 (0.13)	0.12 (0.13)	0.12 (0.13)	0.16 (0.13)	0.15 (0.13)
Log export _{jt}	0.16* (0.10)	0.17* (0.10)	0.19* (0.10)	0.18* (0.10)	0.17* (0.10)	0.18* (0.10)	0.20** (0.08)	0.21** (0.10)	0.22** (0.10)	0.22** (0.10)	0.25** (0.10)	0.25** (0.10)
Constant	-13.87*** (2.82)	-13.11*** (3.30)	-14.74*** (2.74)	-14.74*** (2.74)	-14.07*** (2.60)	-14.24*** (2.72)	-12.47*** (3.40)	-9.43*** (3.50)	-11.11*** (3.34)	-11.13*** (3.33)	-13.61*** (3.50)	-13.61*** (3.56)
Observations	3,813	3,813	3,813	3,813	3,813	3,813	3,407	3,407	3,407	3,407	3,407	3,407

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t). Each column includes an indicator variable for one of the sanction types.

For overall innovation, we find that arms sanctions have the largest negative impact, reducing the total fractional patent count of inventors from sanctioned countries by 39.3%. Arms sanctions include restrictions on the movement of weapons across borders (trade or exchange). Military sanctions, which include suspension of military aid and other measures, also negatively impact innovation. We hypothesize that sanctions related to the defense sector are more strenuous, have stronger enforcement, and likely target innovation in those sectors finding a similar pattern of coefficients for joint fractional patent counts.¹³ Although trade and financial sanctions do not have statistically significant effects on overall innovation, they reduce joint innovation. This is likely due to the types of restrictions that are included in the trade and financial sanctions. Trade sanctions disrupt trade patterns and limit access to exports and/or imports of equipment needed for research or patenting, as well as the flow of intermediate goods (Wang et al. 2025). This would affect innovation in both the target and the sender countries. Financial sanctions include restrictions on financial transactions, including foreign direct investment, loans, and other funding.

One puzzling result is that travel sanctions increase overall and joint innovation, and this increase is statistically significant at 5%. Travel sanctions typically include visa restrictions, which would prevent inventors and researchers from traveling to collaborate in person or attending conferences and events. It would be reasonable to expect that travel sanctions would have a negative impact on innovation due to the restriction of mobility. One potential explanation for our finding is the substitution effect of

¹³We will explore differences in industry level patenting in our future work.

in-person collaboration with virtual collaboration. This is especially plausible with the rise of virtual communication after the onset of the COVID-19 pandemic. To test this hypothesis, we re-estimate equation (1) with the *Travel sanction_{jt}* variable on a sample truncated to pre-COVID-19 period (2000-2019). Although the coefficient of interest for the joint fractional patent count specification is still positive (0.143), the statistical significance disappears.¹⁴ Similarly, for the overall fractional patent count, the coefficient in the variable *Travel sanction_{jt}* is still positive, but the significance level is reduced to 10%. We conclude that travel sanctions encourage innovation in the sender country due to the substitution of in-person collaboration to virtual collaboration, and that this effect is especially prominent after the increase in the usage of virtual communication in the wake of the COVID-19 pandemic.

Table 11: Travel Sanctions: COVID-19 Effect				
	Truncated sample		Variables	
	(1)	(2)	(3)	(4)
	Overall	Joint	Overall	Joint
<i>Travel_{jt}</i>	0.58*	0.14		
	(0.33)	(0.11)		
<i>Travel_{jt} × pre-COVID_t</i>			0.71**	0.17
			(0.29)	(0.11)
<i>Travel_{jt} × post-COVID_t</i>			1.27***	0.62***
			(0.40)	(0.18)
<i>GDP per capita_{jt}</i>	1.69***	1.09***	1.80***	1.08***
	(0.34)	(0.20)	(0.32)	(0.21)
<i>Dipl disagree_{jt}</i>	-0.072*	-0.034	-0.093**	-0.069
	(0.040)	(0.068)	(0.045)	(0.054)
<i>WTO member_{jt}</i>	0.031	-0.29**	-0.11	-0.31***
	(0.22)	(0.13)	(0.25)	(0.12)
<i>EU member_{jt}</i>	0.28*	0.14	-0.044	-0.12*
	(0.16)	(0.16)	(0.064)	(0.062)
<i>Ave edu_{jt}</i>	0.057	0.10	0.094	0.13*
	(0.065)	(0.062)	(0.070)	(0.064)
<i>COV19 restr_{jt}</i>			0.017	0.022
			(0.036)	(0.059)
<i>Log import_{jt}</i>	0.059	0.16	0.034	0.16
	(0.12)	(0.13)	(0.11)	(0.13)
<i>Log export_{jt}</i>	0.18**	0.29***	0.19**	0.27***
	(0.084)	(0.087)	(0.096)	(0.091)
Constant	-13.04***	-14.09***	-14.16***	-13.65***
	(2.92)	(3.42)	(2.59)	(3.48)
Observations	3,309	2,942	3,813	3,407

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t). Columns 1 and 2 estimate equation (1) with an indicator variable for travel sanctions on a sample truncated to pre-COVID-19 period. Columns 3 and 4 re-estimate equation (1) on a full sample, but the impacts of the travel sanction are separated into pre-COVID-19 and post-COVID-19 periods.

We also disentangle the impacts of sanctions based on how innovative country j is. We estimate equation (2).

¹⁴These results are presented in Table 11.

$$y_{jt} = \exp(\alpha + \beta_i \sum_{i=1}^4 Quartile_{ij} \times US\ sanction_{jt} + X_{jt} + \gamma_j + \gamma_t + \epsilon_{jt}) \quad (2)$$

where the dependent variable y_{jt} represents either overall or joint patents: $y_{jt} \in (OVERALLpatent_{jt}, JOINTpatent_{jt})$. Each country is assigned to one of the four quartiles, denoted $Quartile_{ij}$, based on its median innovation throughout the sample. The assignment of the country quartile does not change in the sample. Q1 represents the countries with the lowest level of innovation, and Q4 includes the countries with the highest level of innovation. All other subscripts and variables are the same as previously defined. We estimate equation (2) on the full sample, for both overall and joint innovation. We plot the coefficients for the four quartiles in Figure 6. We also report the p-values of the pairwise comparison tests of the quartile coefficients in Table 12.

Figure 6: Sanction Effects by Innovation Quartile

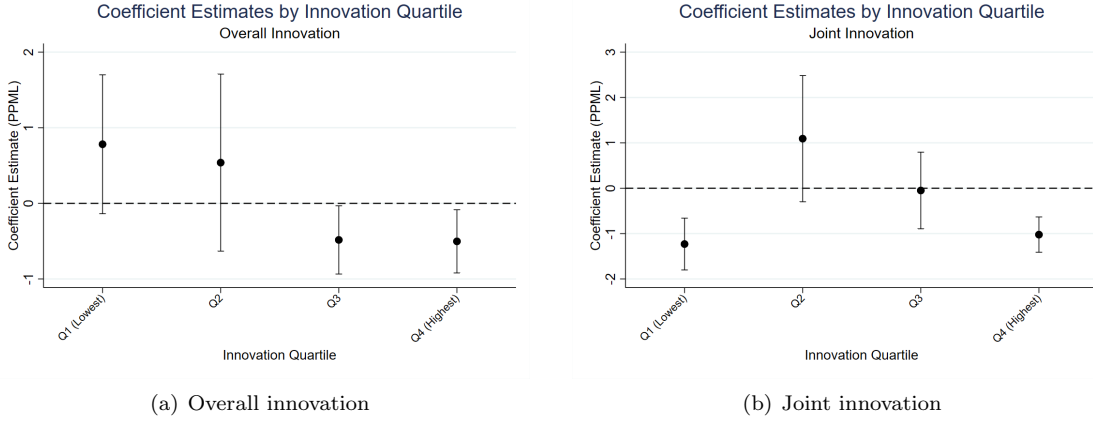


Table 12: Pairwise Comparisons of Innovation Quartile Coefficients

	Overall Innovation			Joint Innovation		
	Q2	Q3	Q4	Q2	Q3	Q4
Q1	0.747	0.015**	0.010**	0.002***	0.017**	0.414
Q2	—	0.115	0.073*	—	0.170	0.002***
Q3	—	—	0.951	—	—	0.035**

Notes: P-values from tests of equality between innovation quartile coefficients. *p<0.1; **p<0.05; ***p<0.01. Q1 = lowest innovation quartile, Q4 = highest innovation quartile.

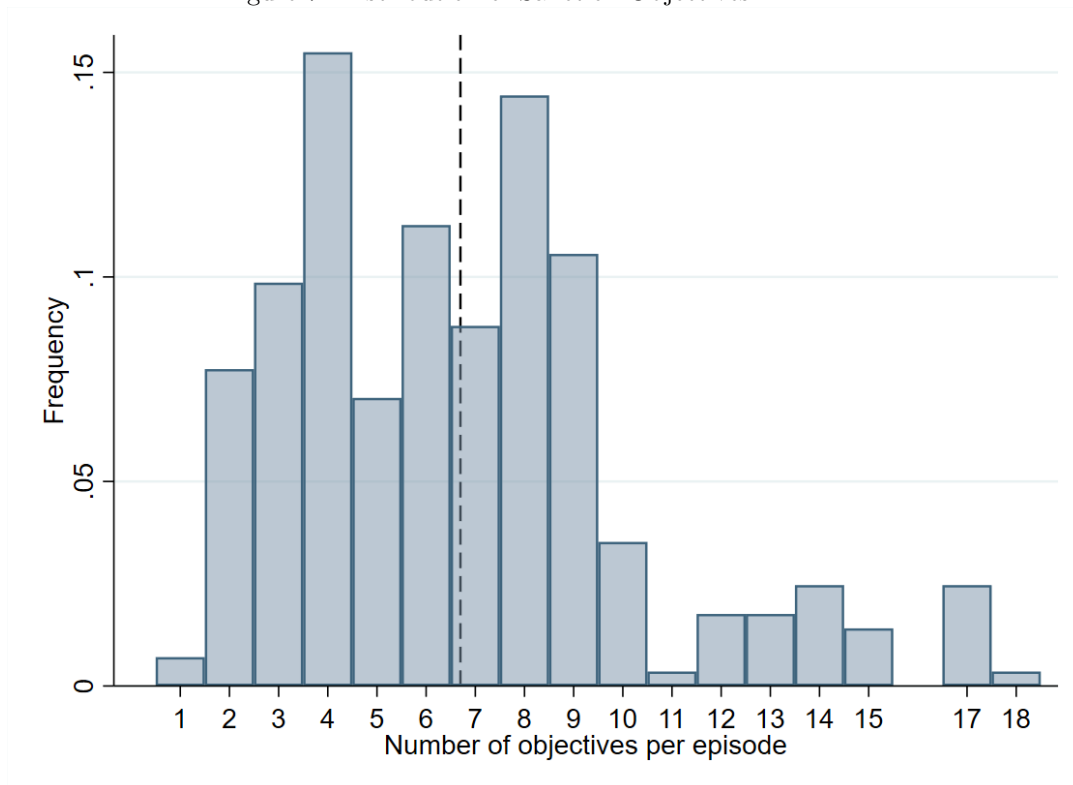
We find that overall innovation decreases in a statistically significant way for inventors from countries in the top two quartiles. The difference between these effects is not statistically significant. For joint innovation, we find statistically significant declines in fractional patent counts for the lowest and highest innovation quartiles. Our findings highlight the heterogeneity in patent responses based on the average level of innovation of the country j .

While a statistically significant decline in joint innovation is intuitive for inventors in countries with

the lowest median patent count, the strong effect for top innovators is surprising. One explanation could be that more innovative countries are more integrated in the US patenting ecosystem and sanctions disturb this integration. This finding could also reflect that innovators from Q4 countries patent more frequently in frontier industries, which sanctions could target (for example, AI, semiconductors, nuclear technology, weaponry).

Finally, we analyze whether the effects of sanctions on patenting differ by the objectives and outcomes of the sanctions. Each sanction case can have multiple objectives. GSDB separates sanctioning objectives into 8 categories: policy change, prevent war, end war, protection of human rights, protection of democracy, combating terrorism, destabilizing the regime, resolution of territorial conflict, and others. There are on average 6 objectives in a sanction case. In Figure 7, we present the distribution of the number of objectives in sanction cases.

Figure 7: Distribution of Sanction Objectives



Notes: Dashed vertical line represents the average number of sanctions objectives, which is equal to 6.7.

The success of sanctions, or their effectiveness, is typically defined by whether the objective was achieved. GSDB distinguishes 5 possible outcomes for sanctions: failed, negotiation settlement, ongoing, partial success, and total success. Thus, each sanction case has multiple objectives, and each objective has a recorded outcome. We use these features of the GSDB to disentangle the impacts of sanctions based on objectives and outcomes.

Table 13: Heterogeneity of Sanction Objectives and Outcomes

	Objective		Success	
	Overall	Joint	Overall	Joint
US sanction _{jt}	-0.54*** (0.18)	-0.52*** (0.19)	-0.44 (0.30)	-0.91*** (0.26)
Objective types:				
US sanction _{jt} × Policy _{jt}	0.08 (0.27)	-0.40* (0.21)		
Success types:				
US sanction _{jt} × Successful _{jt}			0.02 (0.13)	0.16 (0.11)
US sanction _{jt} × Failed _{jt}			-0.10 (0.26)	-0.25 (0.20)
US sanction _{jt} × Ongoing _{jt}			0.14** (0.06)	0.25*** (0.10)
GDP per capita _{jt}	1.82*** (0.33)	1.06*** (0.20)	1.82*** (0.33)	1.06*** (0.20)
Dipl disagree _{jt}	-0.10** (0.05)	-0.06 (0.06)	-0.10** (0.05)	-0.06 (0.06)
WTO member _{jt}	0.46 (0.34)	-0.06 (0.12)	0.46 (0.34)	-0.06 (0.12)
EU member _{jt}	-0.01 (0.08)	-0.09 (0.06)	-0.01 (0.08)	-0.09 (0.06)
Ave edu _{jt}	0.11 (0.08)	0.13** (0.06)	0.11 (0.08)	0.13** (0.06)
COV19 restr _{jt}	0.01 (0.04)	0.02 (0.06)	0.01 (0.04)	0.02 (0.06)
Log import _{jt}	-0.02 (0.13)	0.14 (0.13)	-0.02 (0.13)	0.14 (0.13)
Log export _{jt}	0.17* (0.10)	0.20** (0.08)	0.17* (0.10)	0.20** (0.08)
Constant	-13.88*** (2.85)	-12.34*** (3.45)	-13.86*** (2.85)	-12.30*** (3.46)
Observations	3,813	3,407	3,813	3,407

Notes: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered at the country level. All columns include country fixed effects (γ_j) and year time fixed effects (γ_t).

We begin with the objective analysis. The most widely used objective is policy change. We create an indicator variable *Policy_{jt}*, which is equal to 1 if there is at least one policy change objective in a given sanction case and zero otherwise. We include an interaction between this newly defined indicator and our main variable of interest, *US sanction_{jt}* in equation (1). These results are presented in columns (1) and (2) of Table 13. We find that sanctions with policy change objectives have less of a negative impact on overall fractional patent count, and this difference is statistically significant at 5%. It is the opposite

for joint innovation: sanctions targeted for policy change decrease innovation more than sanctions with other objectives. This could be due to the nature of joint innovation itself, rather than the nature of sanctions. Forming patenting partnerships is costly, and sanctions disrupt these collaborations.

To analyze the impacts of the sanction outcomes, we create three additional indicator variables: *Successful_{jt}* combines partially and completely successful objectives, *Failed_{jt}* takes the value of 1 if there is at least one failed objective, and *Ongoing_{jt}* captures ongoing sanctions. We then interact these outcome indicators with our variable of interest, *US sanction_{jt}*, and include these interactions in equation (1). These results are presented in columns (3) and (4) of Table 13. We do not find a statistically significant difference between successful and ongoing sanctions. They tend to have marginally milder impacts on overall and joint innovation. However, we find that failed sanctions have a more detrimental impact on overall and joint innovation. This could be because once sanctions fail, more serious diplomatic or military actions are taken against the target, which would erode established innovation networks.

We find that US sanctions substantially reduce innovation by inventors from sanctioned countries within the United States and weaken joint collaboration. Strenuous sanctions are associated with a 38% decline in patents filed by sanctioned-country inventors in the US, while joint patenting with US co-inventors falls by roughly 59%. These effects are robust across alternative samples, estimators, and sanction definitions. The effects of sanctions gradually materialize, emerging after four to five years, highlighting the persistent and cumulative nature of the decline in innovation that follows sanctions. We find that the heterogeneity impacts of sanctions is driven by their type, objective, outcome, duration, and strength.

5 Conclusion

The impact of economic sanctions on innovation is a new and evolving area of research. This paper shows how economic sanctions that the US imposes against other countries can backfire by reducing the sanctioned countries patenting in the US, as well as having a detrimental impact on joint innovation.

We find that US sanctions reduce the overall fractional patent count of inventors from sanctioned countries within the United States and weaken joint collaboration. Strenuous sanctions are associated with a 38% decline in patents filed by inventors from sanctioned countries in the US, corresponding to a reduction of approximately 246 patents. Joint patenting between US inventors and collaborators from sanctioned countries drops by about 59%, which in real terms is a decrease of approximately 17 patents. Sanctions tend to have a persistent negative impact on innovation, with negative impacts materializing after 4-5 years. Failed sanctions are more detrimental to innovation, as well as arms and military sanctions. Interestingly, travel sanctions lead to an increase in joint innovation. Our findings are robust across alternative samples, estimators, and sanction definitions.

Future work will examine how these effects vary across industries, shedding light on whether certain sectors are more vulnerable to sanctions-induced innovation losses. We also plan to investigate the relationship between sanctions and knowledge transfer by analyzing patterns in patent citations, which will allow us to assess how sanctions reshape the diffusion of ideas across national boundaries. This emerging line of research can offer valuable guidance to policymakers by highlighting the unintended consequences of sanctions for global innovation and knowledge flows.

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Appendix

Table A1: List of Countries by Innovation Quartile

Q1	Q2	Q3	Q4
Albania	Afghanistan	Armenia	Argentina
Angola	Algeria	Bahamas (the)	Australia
Anguilla	Andorra	Barbados	Austria
Antigua and Barbuda	Azerbaijan	Belarus	Belgium
Belize	Bahrain	Bermuda	Brazil
Benin	Bangladesh	Cayman Islands	Bulgaria
Botswana	Bolivia	Colombia	Cameroon
Burkina Faso	Bosnia and Herzegovina	Costa Rica	Canada
Burundi	Brunei Darussalam	Croatia	Chile
Cabo Verde	Chad	Cuba	China
Cambodia	Democratic Republic of the Congo	Cyprus	Czechia
Central African Republic	El Salvador	Dominican Republic	Denmark
Congo	Ghana	Ecuador	Finland
Côte d'Ivoire	Gibraltar	Egypt	France
Djibouti	Guatemala	Estonia	Germany
Dominica	Guinea-Bissau	Georgia	Greece
Eritrea	Honduras	Iceland	Hungary
Eswatini	Iraq	Indonesia	India
Ethiopia	Isle of Man	Iran	Ireland
Faroe Islands	Jersey	Jamaica	Israel
Fiji	Liberia	Jordan	Italy
Gabon	Moldova	Kazakhstan	Japan
Gambia	Nepal	Kenya	South Korea
Greenland	Oman	North Korea	Luxembourg
Grenada	Paraguay	Kuwait	Malaysia
Guernsey	Republic of North Macedonia	Latvia	Mexico
Guinea	Senegal	Lebanon	Netherlands
Guyana	Tanzania, United Republic of	Liechtenstein	New Zealand
Haiti	Uzbekistan	Lithuania	Norway
Kiribati	Yemen	Malta	Philippines
Kyrgyzstan	Zimbabwe	Mauritius	Poland
Lao People's Democratic Republic		Monaco	Portugal
Lesotho		Morocco	Romania
Libya		Nigeria	Russia
Madagascar		Pakistan	Saudi Arabia
Malawi		Panama	Singapore
Mali		Peru	Slovenia
Micronesia		Qatar	South Africa
Mongolia		Serbia	Spain
Montenegro		Slovakia	Sweden
Mozambique		Sri Lanka	Switzerland
Myanmar		Syrian Arab Republic	Taiwan
Namibia		Trinidad and Tobago	Thailand
Nicaragua		Tunisia	Turkey
Niger		United Arab Emirates	Uganda
Palau		Uruguay	Ukraine
Palestine, State of		Venezuela	United Kingdom
Papua New Guinea		Viet Nam	United States of America
Rwanda		British Virgin Islands	
Saint Helena, Ascension and Tristan da Cunha			
Saint Kitts and Nevis			
Saint Lucia			
Samoa			
San Marino			
Seychelles			
Sierra Leone			
Solomon Islands			
Somalia			
Sudan			
Suriname			
Tajikistan			
Timor-Leste			
Togo			
Turkmenistan			
Turks and Caicos Islands			
Vanuatu			
Zambia			